Fall 1978

Problem 1 (Fa78, Sp12). Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(x) \leqslant f(y)$ for $x \leqslant y$. Prove that the set where f is not continuous is finite or countably infinite.

Problem 2 (Fa78). Let $\{g_n\}$ be a sequence of Riemann integrable functions from $[0, 1]$ into $\mathbb{R}$ such that $|g_n(x)| \leqslant 1$ for all n, x. Define

$$G_n(x) = \int_0^x g_n(t)\,\mathrm{d}t$$

Prove that a subsequence of $\{G_n\}$ converges uniformly.

Problem 3 (Fa78). Let $M_{n\times n}$ denote the vector space of $n \times n$ real matrices. Prove that there are neighborhoods U and V in $M_{n\times n}$ of the identity matrix such that for every A in U, there is a unique X in V such that $X^4 = A$.

Problem 4 (Fa78, Fa09). Evaluate $\displaystyle\int_0^{2\pi} \frac{\mathrm{d}\theta}{1 - 2r\cos\theta + r^2}$ where $|r| \neq 1$.

Problem 5 (Fa78, Su79). Let $f(z) = a_0 + a_1 z + \cdots + a_n z^n$ be a complex polynomial of degree $n > 0$. Prove

$$\frac{1}{2\pi i} \int_{|z|=R} z^{n-1} |f(z)|^2\,\mathrm{d}z = a_0 \bar{a}_n R^{2n}$$

Problem 6 (Fa78). Solve the differential equation $\dfrac{\mathrm{d}y}{\mathrm{d}x} = x^2 y - 3x^2$ for $y(0) = 1$.

Problem 7 (Fa78). Let H be a subgroup of a finite group G.

1. Show that H has the same number of left cosets as right cosets.
2. Let G be the group of symmetries of the square. Find a subgroup H such that xH $\neq$ Hx for some x.

Problem 8 (Fa78, Fa84). Let M be the $n \times n$ matrix over a field $\mathbb{F}$, all of whose entries are equal to 1.

1. Find the characteristic polynomial of M.
2. Is M diagonalizable?

3. Find the Jordan Canonical Form of M. Does the Jordan form depend on the characteristic of the field $\mathbb{F}$.

Problem 9 (Fa78). For $x, y \in \mathbb{C}^n$, let $\langle x, y\rangle$ be the Hermitian inner product $\sum_j x_j \overline{y}_j$. Let T be a linear operator on $\mathbb{C}^n$ such that $\langle Tx, Ty\rangle = 0$ if $\langle x, y\rangle = 0$. Prove that $T = kS$ for some scalar k and some operator S which is unitary: $\langle Sx, Sy\rangle = \langle x, y\rangle$ for all x and y.

Problem 10 (Fa78). How many homomorphisms are there from the group $\mathbb{Z}_2 \times \mathbb{Z}_2$ to the symmetric group on three objects?

Problem 11 (Fa78). Let $W \subset \mathbb{R}^n$ be an open connected set and f a real valued function on W such that all partial derivatives of f are 0. Prove that f is constant.

Problem 12 (Fa78). Let $f : \mathbb{R}^n \to \mathbb{R}$ have the following properties: f is differentiable on $\mathbb{R}^n \setminus \{0\}$, f is continuous at 0, and

$$\lim_{p\to 0} \frac{\partial f}{\partial x_i}(p) = 0$$

for $i = 1, \ldots, n$. Prove that f is differentiable at 0.

Problem 13 (Fa78). Show that if $u, v : \mathbb{R}^2 \to \mathbb{R}$ are continuously differentiable and $\dfrac{\partial u}{\partial y} = \dfrac{\partial v}{\partial x}$, then $u = \dfrac{\partial f}{\partial x}$, $v = \dfrac{\partial f}{\partial y}$ for some $f : \mathbb{R}^2 \to \mathbb{R}$.

Problem 14 (Fa78, Fa10). Give examples of conformal maps as follows:

1. from $\{z \mid |z| < 1\}$ onto $\{z \mid \Re z < 0\}$,
2. from $\{z \mid |z| < 1\}$ onto itself, with $f(0) = 0$ and $f(1/2) = i/2$,
3. from $\{z \mid |z| < 1\}$ onto the sector $\{z \mid 0 < \arg z < \pi/4\}$,
4. from $\{z \mid z \neq 0, 0 < \arg z < 3\pi/2\}$ onto $\{z \mid z \neq 0, 0 < \arg z < \pi/2\}$.

Problem 15 (Fa78, Fa18). Suppose $h(z)$ is analytic in the whole plane, $h(0) = 3 + 4i$, and $|h(z)| \leqslant 5$ if $|z| < 1$. What is $h'(0)$?

Problem 16 (Fa78, Sp79). Which of the following matrices are similar as matrices over $\mathbb{R}$?

$$A_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad A_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad A_3 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$A_4 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \quad A_5 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad A_6 = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Problem 17 (Fa78). Find all automorphisms of the additive group of rational numbers.

Problem 18 (Fa78, Fa85). Prove that every finite multiplicative group of complex numbers is cyclic.

Problem 19 (Fa78). Let

$$A = \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix}$$

Express A^{-1} as a polynomial in A with real coefficients.

Problem 20 (Fa78). Let $M_{n\times n}$ be the vector space of real $n \times n$ matrices, identified with $\mathbb{R}^{n^2}$. Let $X \subset M_{n\times n}$ be a compact set. Let $S \subset \mathbb{C}$ be the set of all numbers that are eigenvalues of at least one element of X. Prove that S is compact.